From FusHa to Folk: Exploring Cross-Lingual Transfer in Arabic Language Models

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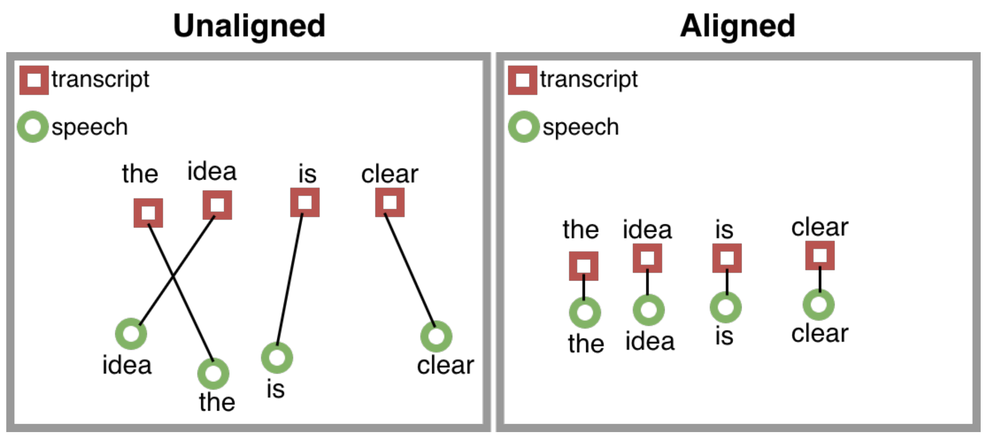
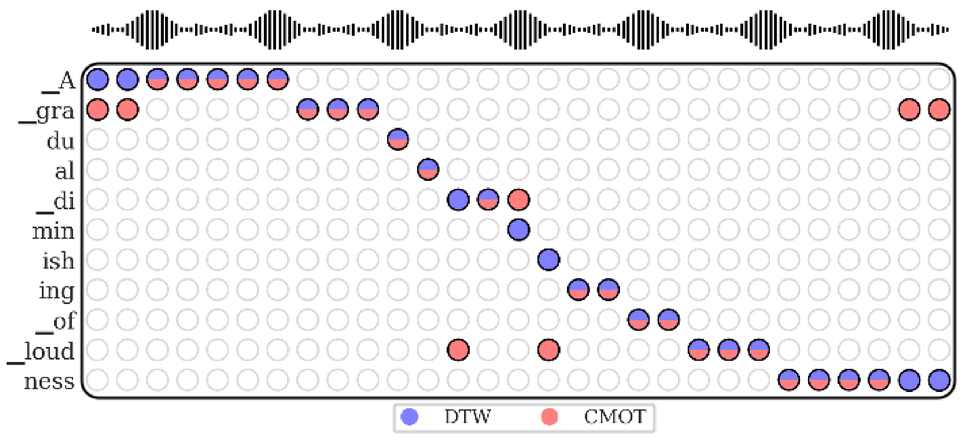
Introduction

How to bridge between the input (speech) and the output (text) modalities in E2E-ST?

- Finding an alignment between speech and text transcriptions.
- Training with an objective that maximizes the alignment.

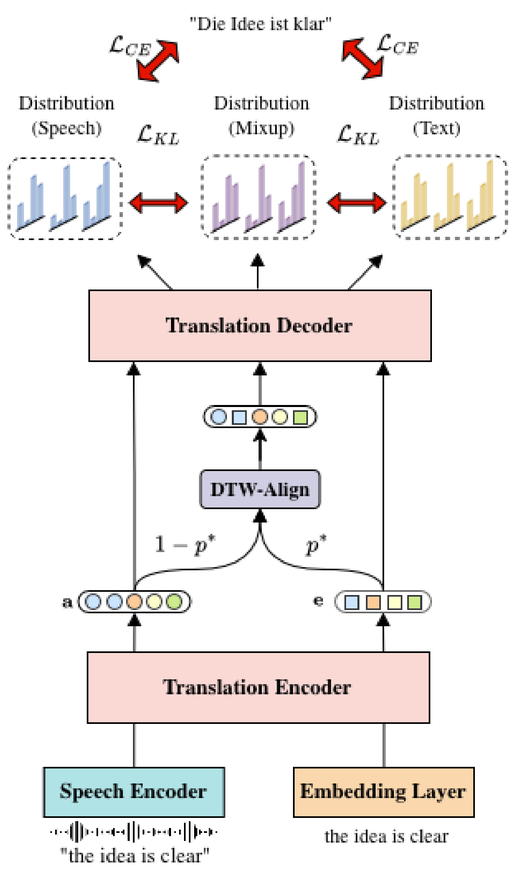


- To find an alignment between speech and text, previous work require an alignment tool but **CMOT** does not.
- We propose **DTW-Align**, which relies on DTW to align speech and text representations during training.
- Unlike CMOT, our method guarantees monotonic alignment and that all tokens are aligned.

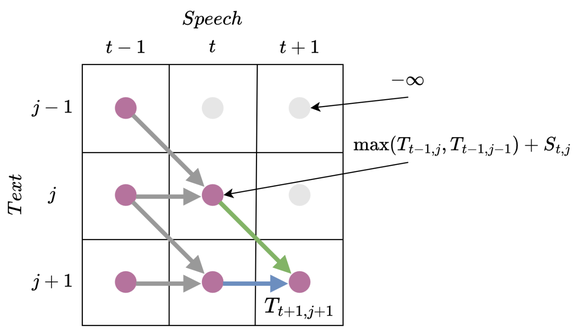


Methodology

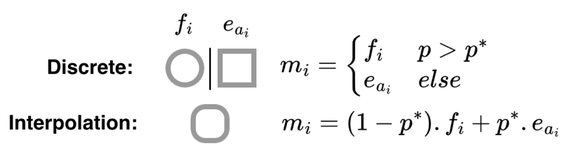
- DTW-Align is used to align speech frames and text tokens.
- The alignment is used for **mixup training**.



DTW-Align:



Discrete vs. Interpolation Mixup:



Main Results

- CMOT, DTW-Align, and DTW-Align-Discrete achieve the best results against other baselines.

Model	En-De	En-Ca	En-Ar	De-En	Fr-En	Es-En	Avg.
Revist-ST †	17.5	22.9	12.3	14.1	26.9	15.7	-
U2TT (Large) †	-	-	-	16.7	27.4	28.1	-
DUB (Large) †	-	-	-	19.5	29.5	**30.9**	-
SRPSE †	-	-	-	21.4	29.3	-	-
CoVoST-2 †	18.4	23.6	13.9	18.9	27.0	28.0	21.6
CTC+OT †	20.6	26.5	15.3	20.4	28.4	29.2	23.4
HuBERT-Transformer	21.4	27.4	15.7	21.8	28.4	28.0	23.8
CMOT	**21.8**	28.2	**16.2**	23.6	30.9	29.6	**25.0**
NFA-Align	21.4	28.1	16.0	23.5	30.9	29.8	24.9
DTW-Align-Discrete	21.7	**28.3**	**16.2**	23.5	**31.0**	29.4	**25.0**
DTW-Align	**21.8**	28.2	16.1	**23.7**	30.8	29.5	**25.0**

†indicates that the results are as reported in the original work.

Alignment Accuracy and Training time

- DTW-Align is significantly faster and more accurate than CMOT.

Method	Accuracy ↑	Execution Time ↓	Train Time ↓
CMOT	26%	97.89	14:20:53
DTW-Align	**45%**	**2.91**	**6:48:14**

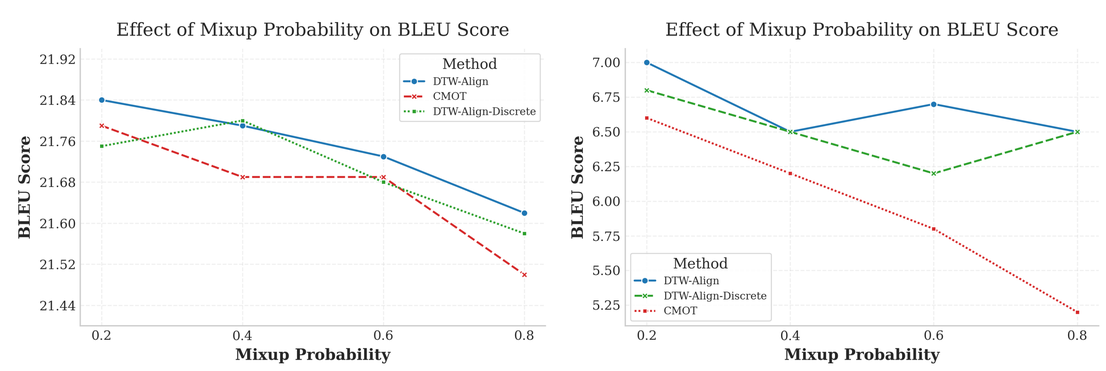
Synthetic Low Resource Setting

- DTW-Align achieves better results in low resource settings.
- Indicates that alignment accuracy is more important in low resource scenarios.
- Shows that interpolation mixup is more robust and data efficient.

Model	En-De	En-Ca	En-Ar	De-En	Fr-En	Es-En	Avg.
HuBERT-Transformer	6.4	8.7	2.2	1.8	3.2	2.9	4.2
CMOT	6.6	9.6	2.7	2.8	8.5	7.5	6.3
DTW-Align-Discrete	6.8**	9.6	2.8	2.7	**8.6**	7.9**	6.4
DTW-Align	**7.0****	**9.8****	**3.1****	**2.9***	**8.6**	**8.0****	**6.6**

Effect of the Mixup Probability

- Increasing the mixup probability p^* (more text) hurts CMOT more.
- This is an implication of lower alignment accuracy.



(a) En-De (b) En-De Low Resource

Conclusion

1. We introduce DTW-Align, a method that **combines** DTW for alignment and interpolation mixup.
2. DTW-Align achieves **similar results to CMOT** on 6 language directions.
3. DTW-Align is **faster and more accurate** than CMOT for alignment.
4. In a **low resource setting**, DTW-Align achieves **better results** than CMOT on all language directions.

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